Equal-scale factorization and the bias and torsion joins

Author proposal: /root/astra_v14_baseline_compiler

Protocol 1.4; separate successor to the one-variable D/F packet

This packet does not revise the frozen D/F or Section 13 proofs. It supplies a proposed connecting proof for the previously open D.5-to-E.11/E.13 dependency. All acceptance and incorporation decisions remain independent. The only quantitative equidistribution input is the *final published* equal-side theorem in [GT14, pp.1176–1177], with the published minor corrections to [GT12]. No unpublished arbitrary-side extension is imported.

1 Conventions and exact statements

Fix standard positive integers k, d and an unbounded integer N. All groups, rational structures, lattices, coordinate bases and finite dimensions are standard, with natural ultrapower extensions. A nilpotent Lie group is connected and simply connected. The rational degree filtration $G_\bullet$ has finite length; the source $\mathbb{Z}^k$ has its total-degree filtration. Multiindices have nonnegative integer entries, and $|\alpha| = \sum_i \alpha_i$. All vertical identities for a limit nilcharacter hold at every limit group element and point, as in the accepted tautification-003 convention. We do not substitute equivariance only at standard points.

In fixed Mal'cev coordinates, small polynomial coefficients mean bounded constant coefficient and coefficient $O(N^{-|\alpha|})$ at positive index α in either the monomial or product-binomial basis. The constants are standard and may depend on the actual fixed geometric data and orbit. Total equidistribution on $[N]^k$ means equidistribution on every product of dense arithmetic progressions in $[N]$, with standard Lipschitz tests. It also applies to internal tests with a common standard Lipschitz bound: finite uniform nets in a compact Lipschitz ball reduce those tests to standard ones, with the desired accuracy chosen before taking the limit.

Theorem 1 (Equal-scale factorization, MF-D4)**.** *For $g \in {}^*\mathrm{poly}(\mathbb{Z}^k, G_\bullet)$ there are a standard connected rational subgroup $P \le G$ and internal polynomial maps ϵ, p, γ such that*

$$g(n) = \epsilon(n)p(n)\gamma(n) \qquad (n \in {}^*\mathbb{Z}^k). \tag{1}$$

The map p lies in P with its induced filtration, and is totally equidistributed on $[N]^k$ in P/Δ for every standard finite-index $\Delta \le P \cap \Gamma$, with the same p and P. The coefficients of ϵ are small as specified above; it is bounded and $O(1/N)$-Lipschitz on every fixed standard dilate $[-AN, AN]^k$. Its group Taylor coefficient at positive index α is $O(N^{-|\alpha|})$ as well. There is a standard discrete subgroup $\Lambda \ge \Gamma$ containing all values of γ, and a standard $q > 0$ such that

$$\gamma(n + qe_i)\Gamma = \gamma(n)\Gamma \qquad (n \in {}^*\mathbb{Z}^k, 1 \le i \le k). \tag{2}$$

There is no properness requirement on P, and no claim that γ is Γ-valued. All factors in (1) are defined on the full integer lattice.

Write $I = [-N, N] \cap {}^*\mathbb{Z}$.

Theorem 2 (Actual bias and torsion outputs, MF-E11/E13)**.** *Let $d \geq 2$ be standard, and let χ be a globally represented unit vector nilcharacter of degree d on ${}^*\mathbb{Z}^k$. If a scalar coordinate of χ has nonzero correlation with a degree at most $d-1$ nilsequence after multiplication by the indicator of a convex polytope in I^k and a product of dense progressions in $[N]$, then χ agrees on all of I^k with a globally represented degree-at-most-$(d-1)$ vector nilsequence. The same conclusion holds if $\chi^{\otimes a}$ agrees on I^k with a degree-at-most-$(d-1)$ vector nilsequence for a standard integer $a \geq 1$.*

These assertions give exactly the lower-degree conclusions of [GTZ, E.11] and [GTZ, E.13] on their required symmetric box. They include finite scalar-coordinate reductions for vector correlations. They do not require an extrapolation from an unrelated subprogression or the deletion of a rational Taylor coefficient.

2 A weighted inverse-affine estimate

The following estimate also supplies the large-offset transition behind [GT12, Proposition 9.2]. Merely inserting an offset of size N into the height bound of its Lemma 8.4 would not give the required standard bound.

Lemma 3 (Large offsets, bounded steps, MF-AFFINE)**.** *Let $f(n)$ be a real polynomial of fixed total degree at most D in k variables. Let A_i be internal integers with $|A_i| = O(N)$ and let b_i be nonzero integers of bounded absolute value. Put $\widetilde{f}(m) = f(A_1 + b_1 m_1, \dots, A_k + b_k m_k)$. Suppose every positive product-binomial coefficient $\widetilde{f}_\beta$ satisfies $\|\widetilde{f}_\beta\|_{\mathbb{T}} = O(N^{-|\beta|})$. There is a standard positive integer r, depending only on D, k and the step bounds, such that every positive coefficient of rf has the same bound $O(N^{-|\alpha|})$ modulo one.*

Proof. Invert the substitution over $\mathbb{R}$ and expand

$$\prod_i \binom{(n_i - A_i)/b_i}{\beta_i} = \sum_{\alpha \leq \beta} c_{\alpha\beta} \binom{n}{\alpha}. \tag{3}$$

Expanding each falling factorial first in monomials shows that the denominators divide a fixed integer formed from $\beta_i!$ and $b_i^{\beta_i}$. Conversion of ordinary powers to binomial coefficients uses integer Stirling coefficients. Thus one standard r clears all denominators, independently of the integer numerators A_i. The coefficient of n^ν before the last conversion is a polynomial in the A_i of total degree at most $|\beta| - |\nu|$. Consequently

$$|c_{\alpha\beta}| = O(N^{|\beta|-|\alpha|}), \qquad r c_{\alpha\beta} \in {}^*\mathbb{Z}. \tag{4}$$

The case $\beta = 0$ contributes only to $\alpha = 0$. For each $\beta > 0$ write $\widetilde{f}_\beta = z_\beta + e_\beta$, where z_β is an integer and $e_\beta = O(N^{-|\beta|})$. In the coefficient of index $\alpha > 0$ in rf, the contributions $r c_{\alpha\beta} z_\beta$ are integers, while

$$\sum_{\beta \geq \alpha} r c_{\alpha\beta} e_\beta = O(N^{-|\alpha|}). \tag{5}$$

There are only a fixed number of indices. This proves the claim. Negative steps cause no change after taking a positive common denominator. The uncontrolled constant coefficient causes no positive-index error. □

3 From a dense rectangle to the published equal-side theorem

Lemma 4 (Weak factorization in finitely many parameters)**.** *Suppose $G_0 = G_1 = G$. If $g\Gamma$ is not totally equidistributed on $[N]^k$, it has a factorization (1) with p in a standard proper connected rational subgroup, retaining all conclusions of Theorem 1 except the final equidistribution condition.*

Proof. Failure supplies a product of progressions $P_i = \{a_i + b_i t : 1 \le t \le L_i\} \subseteq [N]$ with $L_i \ge \delta N$, standard $\delta > 0$, and a bounded Lipschitz test with noninfinitesimal discrepancy. Their integer steps obey $1 \le b_i \le 2/\delta$ and their offsets are $O(N)$. Subtract the Haar mean, normalize the test and rotate a constant complex phase so that the real mean discrepancy is positive.

In the index rectangle $\prod[L_i]$, partition each axis into blocks of a common length $L = \lfloor \eta N \rfloor$, leaving a final remainder of length less than L. Choose the standard $\eta > 0$ small compared with the discrepancy, δ, k and the test norm. The discarded proportion is at most $\sum_i L/L_i \le k\eta/\delta + o(1)$, so its contribution is less than one quarter of that discrepancy. At least one full cube of side L has a positive standard discrepancy. On this cube the polynomial is $\widetilde{g}(m) = g(A_1 + b_1 m_1, \dots, A_k + b_k m_k)$ with $|A_i| = O(N)$.

Apply the exact equal-side theorem of [GT14, pp.1176–1177] at length L, using a standard error parameter below this discrepancy and below the reciprocal rationality bound of the fixed adapted basis. The result gives a nonzero bounded horizontal integer character ξ for which all positive coefficients of the real horizontal lift of $\xi\widetilde{g}$ have circle norm $O(L^{-|\alpha|}) = O(N^{-|\alpha|})$. Only finitely many such characters are possible, so the selected character is standard. Lemma 3 replaces it by a nonzero standard multiple and gives the same coefficient bounds for ξg on the original variables. This is the precise rectangle-to-equal-cube join; the original unequal-side theorem is not used.

Normalize $g(0) = c\lambda$ with bounded c and $\lambda \in {}^*\Gamma$ and apply the preceding argument to $c^{-1}g\lambda^{-1}$, whose failure is preserved by bounded left translation. Its coordinates are $\sum_{\alpha>0} t_\alpha \binom{n}{\alpha}$, with t_α supported in $G_{|\alpha|}$ by [GT12, Lemma 6.7]. Write the real horizontal lift as $u \cdot \psi$, with u a fixed integer vector. Choose integers a_α nearest to $u \cdot t_\alpha$. On a support where u is nonzero choose a fixed rational vector v_α of that support with $u \cdot v_\alpha = 1$, and define

$$w_\alpha = (u \cdot t_\alpha - a_\alpha)v_\alpha, \quad z_\alpha = a_\alpha v_\alpha, \quad \psi(\epsilon_0(n)) = \sum w_\alpha \binom{n}{\alpha}, \quad \psi(\gamma_0(n)) = \sum z_\alpha \binom{n}{\alpha}. \tag{6}$$

If u vanishes on a support, set all these vectors and its integer to zero. Coordinate support and polynomial group closure make these maps polynomial for the original filtration. The real lift is identically zero on $p = \epsilon_0^{-1} c^{-1} g \lambda^{-1} \gamma_0^{-1}$. Thus p lies in its standard proper connected rational kernel. The original factors are $\epsilon = c\epsilon_0$ and $\gamma = \gamma_0\lambda$. The w_α have precisely the required small bounds; the z_α have a common standard denominator, without a numerator bound. Controlled products and periods are checked next. □

4 Controlled products, periods, and descent

Scale all variables by $n_i = Nt_i$. Fixed coordinate polynomials with small coefficients become polynomials in t with bounded coefficients. The fixed BCH and coordinate-change polynomials preserve this property under finite products, inverses and fixed rational subgroup embeddings. Conversion between monomial and product-binomial bases is triangular in componentwise indices, so the coefficient of $\binom{n}{\alpha}$ is $O(N^{-|\alpha|})$. Differentiating the scaled coordinate maps on the compact box $[-A, A]^k$ gives the claimed $O(1/N)$ Lipschitz bound on every fixed dilate. A real-coordinate polynomial extension of ϵ has the same bounds, which will be used below.

These estimates also imply the group Taylor bounds. Assign formal weight $|\alpha|$ to each logarithmic coefficient of n^α and weight zero to the bounded constant term. Group Taylor extraction in any fixed admissible order uses fixed BCH polynomials. Inductively every term of formal weight w has coordinate polynomial total degree at most w; removing a Taylor factor at index β preserves this property because its binomial power has degree $|\beta|$. The coefficient at index α cannot contain a term of formal weight less than $|\alpha|$: such a term has total degree less than $|\alpha|$ and has zero coefficient at that index. This is the same triangular Taylor extraction as [GTZ, B.9], now with product-binomial indices. Each surviving BCH monomial contributes at least $N^{-|\alpha|}$ decay. In particular no estimate is inferred solely from separate degree in individual variables.

The common discrete overgroup argument is independent of the number of parameters. In a rational lower-central basis of weights w_i, choose a standard D clearing every coordinate denominator and every fixed BCH coefficient. As in DF002, the additive lattice $\sum_i D^{-(2w_i-1)}\mathbb{Z}X_i$ is BCH-closed: a nonlinear term with t inputs has output denominator exponent exceeding the input sum by at least $t-1$. Its exponential is a standard discrete overgroup $\Lambda \geq \Gamma$ containing the rational factors and any finite product of them.

Here is the full multivariable period argument, in the right-coset direction. Each coordinate of $\gamma(n)$ is a polynomial of fixed degree whose values have a common standard denominator. Tensor-product interpolation on a fixed integer grid gives a standard common denominator for all its coefficients. For every i the vector

$$R_i(n,r) = \psi\big(\gamma(n)^{-1}\gamma(n+re_i)\big)$$

is a rational coefficient polynomial with a common standard denominator. It is identically zero when $r=0$. Choose a standard positive q divisible by all these denominators. Each monomial then has a positive power of r, so $R_i(n,q)$ is integral for every internal integer vector n. This is exactly (2). This proof does not use the wrongly oriented quotient in the printed proof of [GT12, A.12]. The corresponding one-variable sublemma of DF002 has a separate independent root review.

To finish Theorem 1, first normalize the constant term and restrict to G_1 if $G_1 \neq G$. The identity

$$c^{-1}g(n)\lambda^{-1} = \lambda\big(g(0)^{-1}g(n)\big)\lambda^{-1} \in G_1$$

holds in any number of variables, since first derivatives lie in G_1 and the positive groups are normal in G_0. Thus a controlled representation exists with a taut positive filtration. Choose a standard rational representing subgroup of least dimension. For any standard finite-index lattice $\Delta \leq P \cap \Gamma$, failure of total equidistribution in P/Δ permits Lemma 4, after the same normalization if its induced first group is proper. It supplies $p = \epsilon_2 p_2 \gamma_2$ in a proper rational $P_2 < P$. The correct new identity is

$$g = (\epsilon\epsilon_2)p_2(\gamma_2\gamma).$$

The preceding controlled-product and discrete-overgroup arguments verify every required bound and period. Rationality relative to Δ is the same rational structure as relative to $P \cap \Gamma$. This contradicts minimal dimension. It proves total equidistribution for every standard finite-index cover with the same factors. Only a minimum among $0, \ldots, \dim G$ was selected; no infinite sequence of refinements or dependence of the chosen factors on the later cover is hidden here.

5 One quotient for all periodic residue classes

We now work in a taut ambient degree-d group G, so G_d is central. Tautification-003, Theorem 1, supplies this exact representation for each nilsequence used below with the same positive-degree groups and the same standard vertical character. Products of these representations remain taut.

Apply Theorem 1, and choose a standard representative $\lambda_\rho \in \Lambda$ for each value of the right coset of $\gamma(n)$ on a residue class $n \equiv \rho \pmod q$, $\rho \in (\mathbb{Z}/q\mathbb{Z})^k$. Such representatives can be standard: Λ/Γ is a fixed finite set, and periodicity makes the coset constant on each class. Let

$$\Gamma_0 = \bigcap_{\lambda \in \Lambda} \lambda \Gamma \lambda^{-1}, \qquad \Delta = P \cap \Gamma_0. \tag{7}$$

The first intersection is the kernel of the action on the finite coset set, so it is a standard normal finite-index subgroup of Λ contained in Γ. Thus Δ is finite index in $P \cap \Gamma$, and p is totally equidistributed on P/Δ. For every ρ the function

$$x \longmapsto F(zx\lambda_\rho \Gamma) \tag{8}$$

is well-defined on this *same* quotient: for $\delta \in \Delta$, $\lambda_\rho^{-1}\delta\lambda_\rho \in \Gamma$. No deletion of a rational horizontal coefficient is involved.

Lemma 5 (A nonzero integral from the actual cutoff)**.** *Let F be an internal bounded Lipschitz scalar test of standard norm on G/Γ. If*

$$|\mathbb{E}_{n \in I^k} 1_\Omega(n) 1_{P_1 \times \cdots \times P_k}(n) F(g(n)\Gamma)| > c > 0 \tag{9}$$

for a convex polytope $\Omega \subseteq I^k$ and dense progressions $P_i \subseteq [N]$, then for some residue ρ and bounded $z = \epsilon(n_0)$ one has

$$\left| \int_{P/\Delta} F(zx\lambda_\rho \Gamma)\, d\mu(x) \right| > c' > 0. \tag{10}$$

Here c' is standard and positive. The factors and quotient are those just constructed for the same actual g.

Proof. Partition the progression product into its finitely many nonempty residue classes modulo q. Each intersection is again a product of dense progressions, by the elementary congruence calculation for each axis. Their steps are bounded by the old step times q. Rotate a common complex phase in (9); one class retains a real sum at least $c_0 N^k$ for a standard $c_0 > 0$. Fix this lower bound before selecting a mesh.

Use index coordinates on that progression product. The inverse image of Ω intersected with the index rectangle is convex and lies in a box of side $O(N)$. Partition it by cubes of side $\lfloor \eta N \rfloor$. The cubes meeting its boundary have at most $O_k(\eta N^k) + O_k(N^{k-1})$ integer points, with constants also depending on the fixed step bounds. Indeed place disjoint unit cubes at those integer points and enclose them in the $O_k(\eta N + 1)$ neighborhood of the convex boundary; apply the published boundary-volume estimate [GT10, Corollary A.2, p.1810]. For a convex set with empty interior, first thicken by an arbitrarily small ball and then take the decreasing limit of its boundary neighborhoods. Choose standard η small enough that the discarded boundary contribution is below $c_0 N^k/4$. The total number of points in the full cubes is at most N^k. Some cube contained in the convex set therefore gives a dense progression product R whose phase-adjusted average is at least $3c_0/4$.

The choice of η can also ensure that $|F(\epsilon(n)x\lambda_\rho\Gamma) - F(\epsilon(n_0)x\lambda_\rho\Gamma)|$ is below $c_0/4$ for all $n, n_0 \in R$ and all $x \in P/\Delta$. This follows from the $O(1/N)$ variation of ϵ, its bounded image, the fixed standard representative λ_ρ, and compactness of P/Δ. Choose the mesh only after these standard constants and the original positive discrepancy are known. For $n \in R$, (1) and the fixed right coset give $F(g(n)\Gamma) = F(\epsilon(n)p(n)\lambda_\rho\Gamma)$. Freeze ϵ at n_0. Total equidistribution of the same p in P/Δ on R identifies the resulting average with the integral in (10), up to an infinitesimal error. Internal-test uniformity follows from the finite Lipschitz net specified at the start. This proves (10). □

6 Exact reconstruction on the symmetric box

Let $G = G^{\chi} \times G^{0}$ be the taut product of a degree-d nilcharacter representation and any degree-at-most-$(d-1)$ auxiliary representation. The latter has trivial degree-d group. Write F_{χ} for the original unit vector test, and let η be its real standard vertical character. Set

$$K = P_d = P \cap G_d.$$

This is connected, rational and central. These facts follow by taking Lie algebra intersections in a simply connected nilpotent group; the rational subgroup and quotient assertions are precisely the geometry established in LP13-002, Section 2. Since Δ has finite index, K is rational for Δ too. Thus $\overline{P} = P/K$ with lattice $\overline{\Delta}$ is a fixed standard filtered nilmanifold of degree at most $d-1$.

Suppose that

$$\eta(\mathrm{pr}_{\chi} K) = 0. \tag{11}$$

For each ρ and bounded real-coordinate parameter $t \in [-1/5, 1/5]^k$, define, initially on P/Δ,

$$H_{\rho}(x,t) = F_{\chi}(\widetilde{\epsilon}_{\chi}(10Nt)\, x_{\chi}\, (\lambda_{\rho})_{\chi} \Gamma_{\chi}), \tag{12}$$

where $\widetilde{\epsilon}$ is the real-coordinate polynomial extension. This is well-defined by (7). It is invariant under K by the all-limit vertical identity and (11); centrality permits its element to pass across $\widetilde{\epsilon}$ and λ_{ρ}. Hence (12) descends to $\overline{P}/\overline{\Delta}$. The scaled-coefficient estimates imply a standard Lipschitz bound in t, uniformly in N. Smooth multiplication on fixed compact ranges gives the same conclusion in x. Invariance descends this bound to the quotient metric; any fixed standard Riemannian metric is equivalent on this compact nilmanifold.

Choose a fixed Lipschitz function $B(t)$ on $\mathbb{T}^k$ equal to one on $[-1/10, 1/10]^k$ and zero outside $(-1/5, 1/5)^k$. Using centered representatives inside that chart, define $\widetilde{H}_{\rho}(x,t) = B(t) H_{\rho}(x,t)$ there and zero outside. It is a globally bounded Lipschitz vector function. The zero collar makes the definition independent of the chart boundary. The functions are limit functions with a standard bound, since their coordinate-level construction has uniform bounds before taking the ultrapower.

For each $\rho \in (\mathbb{Z}/q\mathbb{Z})^k$ choose a standard Lipschitz function b_{ρ} on $\mathbb{T}^k$ with $b_{\rho}(\sigma/q) = 1_{\rho=\sigma}$. For example use products of triangular bumps of radius $1/q$; for $q = 1$ use the constant function one. The vector function

$$\mathcal{H}(x,t,z) = \sum_{\rho} b_{\rho}(z) \widetilde{H}_{\rho}(x,t) \tag{13}$$

is globally bounded Lipschitz with a standard bound. Its orbit is

$$n \longmapsto (\overline{p}(n)\overline{\Delta}, n/(10N),\, n/q). \tag{14}$$

This orbit has degree at most $d-1$: its first component has that degree and both tori have degree one, with $1 \le d-1$. For every $n \in I^k$, exactly the correct residue value is sampled in (13), and $B(n/(10N)) = 1$. The identities (1), (2), and (12) give

$$\chi(n) = \mathcal{H}((14)). \tag{15}$$

This is an equality of the actual vector functions on the entire symmetric box. It is not an extrapolation of an equality on just one progression. No unit condition is imposed on $\mathcal{H}$ away from the orbit; a lower-degree vector nilsequence does not require one.

7 Bias and torsion applications on the same objects

Proof of Theorem 2. For the bias assertion, choose the scalar coordinate and the lower-degree scalar test occurring in the actual nonzero correlation. If a vector correlation is given, a fixed finite coordinate choice supplies one such scalar nonzero average; its density loss is standard. Form the taut product representation $G^{\chi} \times G^{0}$ and the product test $F = (F_{\chi})_i \overline{F_0}$, using the sign in that correlation. It has vertical character η on the degree-d group because G_d^0 is trivial. Apply Theorem 1 and Lemma 5 to this actual product orbit and test, obtaining a nonzero integral (10).

Translation of Haar measure on P/Δ by any $u \in K$ preserves that integral. The integrand is multiplied by $e(\eta(\mathrm{pr}_{\chi} u))$, since K is central in the ambient taut group. The integral is nonzero, so this phase is one for every standard $u \in K$. The real homomorphism η is continuous and K is connected; its image is therefore a connected subset of $\mathbb{Z}$, hence zero. Its natural ultrapower extension vanishes on every limit element too. This is (11). Equations (12)–(15), with the *original* vector test F_{χ}, now prove the asserted all-box lower-degree representation.

For the torsion assertion, let Ψ be the given degree-at-most-$(d-1)$ vector nilsequence equal to $\chi^{\otimes a}$ on I^k. Put its finitely many coordinates into one common product nilmanifold of that degree. Since χ is a unit vector, pointwise on $[N]^k$ one has

$$\sum_j (\chi^{\otimes a})_j(n)\overline{\Psi_j(n)} = 1.$$

At least one scalar term has a nonzero standard average, with lower bound at least the reciprocal of the fixed tensor dimension in absolute value. Use that term as the test on the product of the original χ geometry and the auxiliary geometry of Ψ. Its top vertical character is $a\eta$. Lemma 5 applies with the full positive cube and no nontrivial cutoff. The same Haar argument gives $a\eta(\mathrm{pr}_{\chi} K) = 0$ as a real homomorphism. As $a \geq 1$, (11) follows. The same reconstruction (12)–(15) uses the original F_{χ}, not its tensor power, and proves the required conclusion.

In the symbol formulation of [GTZ, E.13], $a[\chi] = 0$ means exactly that the tensor power is equivalent to the constant nilcharacter. By the definition of equivalence, its coordinates are lower-degree nilsequences on the specified box. This is precisely the Ψ premise just used, with a finite common representation. No torsion-free conclusion in degree one is asserted; the time and residue torus factors explain the requirement $d \geq 2$. □

8 Boundaries of this proposal

This proof supplies the finite-parameter factorization actually needed by E.11/E.13 and their all-symmetric-box reconstruction. It does not assert the false horizontal Taylor closeness printed in D.5. Gaussian's separate one-variable D.6 argument must retain its denominator correction in the degree-rank horizontal quotient. The inverse-affine Lemma 3 also validates the large-offset coefficient transition in the one-variable weak-factorization source. That is a related use requiring an explicit link to this frozen version; the earlier DF002 bytes and earlier decisions are preserved. The Section 13 application still needs an independent decision binding its exact E.11/E.13 objects and domains to this proposal, and the complete inverse theorem still needs its other separately assigned components.

References

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